DATA CHART GUIDE FOR FLIGHT

VIPERFISH OPERATORS KEY TOOL

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1 Introduction

This document gives a ViperFish operator a guide of how to analyse the **Data Chart** window in the Flight software. Understanding the **Data Chart** window will help the operator to make changes to the system that will give the ViperFish the best possible circumstances for being a stable platform with optimal manoeuvrability.

Note: For operators familiar with using a ScanFish with Flight, the flap angles are opposite for the ViperFish.

2 Data Chart introduction

The **Data Chart** window is opened from within Flight via **View** > **Dockable Panels** > **Data Chart**.

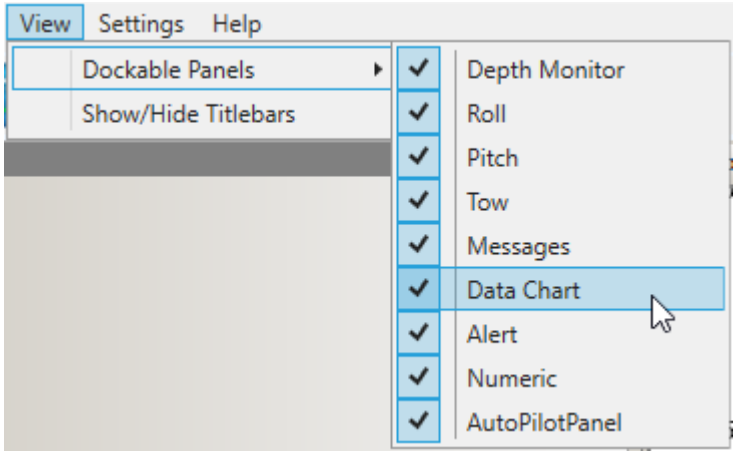

Figure 1 How to open a Data Chart window in Flight

The tick mark to the left of **Data Chart** marks that one instance of the Data Chart is already open. It is possible to open multiple **Data Chart** windows, unlike many of the other windows in Flight which can be opened once from this menu.

The window itself is a graph window where numerical values (y-axis) can be displayed as a function of time (x-axis).

The available numerical values that can be displayed are:

- **2D Flap Angle Port**: Displays the flap angles for port flap
- **2D Flap Angle Stbd**: Displays the flap angles for starboard flap
- **2D Flap Command Port**: Displays the command angle sent to the port flap
- **2D Flap Command Stbd**: Displays the command angle sent to the starboard flap
- **3D Flap Angle Port**: Displays the flap angles for port 3D flap
- **3D Flap Angle Stbd**: Displays the flap angles for starboard 3D flap
- **Altimeter**: Displays the altitude of the ViperFish
- **Battery Voltage**: Displays the ADC of the ViperFish
- **Cable Counter Speed**: Displays the speed in m/s of the winch cable
- **Cable Counter Value**: Displays the catenary length of winch cable
- **Controller Temperature**: Displays the hardware temperature in the controller
- **Distance Off Line**: Displays the cross track error in metres
- **Heading**: Displays the ViperFish heading
- **Pitch**: Displays the ViperFish pitch
- **Roll**: Displays the ViperFish roll
- **Water Depth**: Displays the depth of the ViperFish

Combinations of these different numerical values can be arranged in the different open Data Chart windows.

Tip: A generally recommended combination to have open is the both the port and starboard 2D flap motors' command and actual flap motor angles, as shown in Figure 2 below:

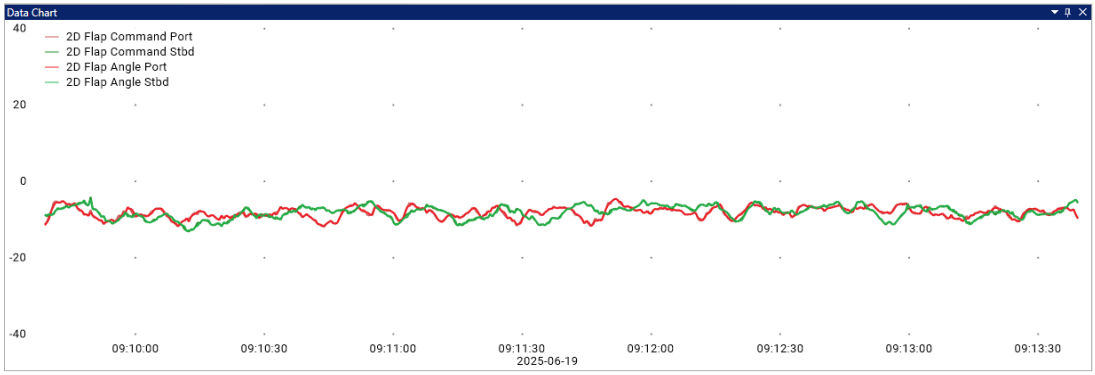

Figure 2 Data Chart combination of both flap motors' command and actual motor angles that is recommended to always have opened when operating the ViperFish

It is possible to make changes to the **Data Chart** window from **Settings** > **Option** > **User Interface** > **Data Chart Panel**. From here, the pen size and colour of each numerical value that can be displayed in the **Data Chart** can be changed.

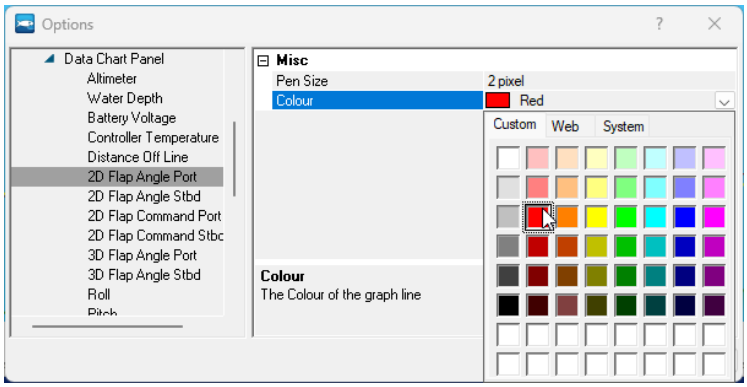

Figure 3 How to change graph colour in the Data Chart

Note: For the port and starboard flaps, it is recommended to have *red* and *green* colours, respectively.

Selecting the **Data Chart Panel** entry, under the **Colours** heading the **Grid Colour** and **Background Colour** can be changed; under **Value Axis** the **Value-Axis Mode, Minimum Value** and **Maximum Value** can be set; and under **Misc** the **Data Age** and **Legend Location** can be set, as shown in Figure 4.

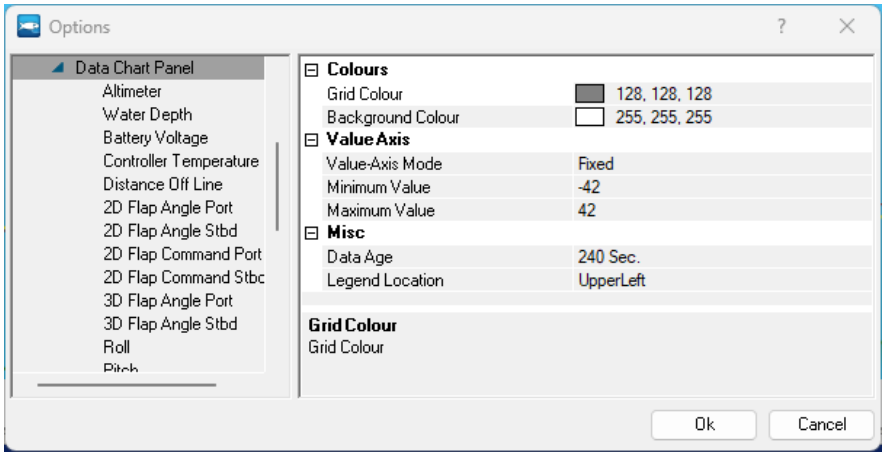

Figure 4 Settings for the Data Chart panel

The **Value-Axis mode** has two different options to choose from; **Fixed** and **Auto**.

It is recommended to set the mode to **Fixed** for the flap motors, and the **Minimum Value** set to *-42* and **Maximum Value** set to *42*.

The motors are limited to go ±40 degrees during operation to prevent physical contact between the grub screw and end stop of the motor (read the ViperFish manual for more information on this at EIVA FreshDesk). By setting the **Minimum Value** and **Maximum Value** to 2 degrees more, the operator will be able to see when the motor is at its outer-

most position in either of the two directions (as in Figure 5, where the port motor is locked out at constant 40° for the last ~45 seconds).

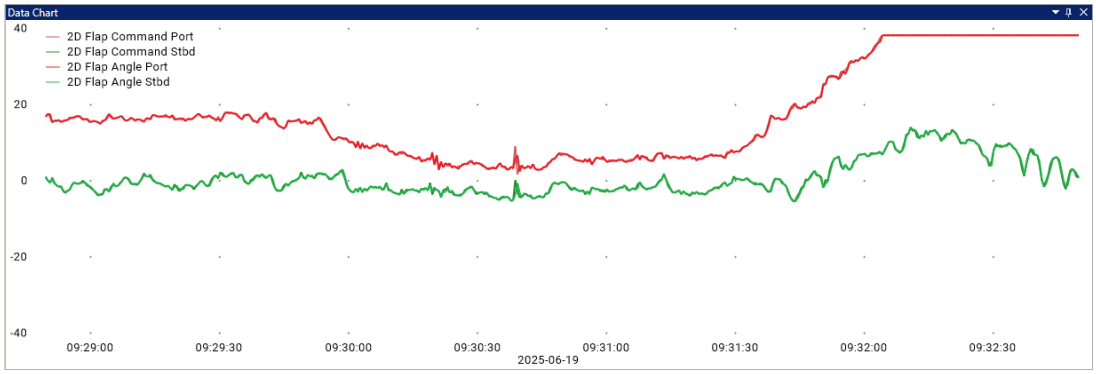

Figure 5 The port flap motor is at the maximum flap angle value

It is recommended to have the same data age in the **Data Chart** as in the **Depth Monitor** window, to easily compare flap changes with seabed changes between the two.

3 Making adjustments based on observations in Data Chart

A perfectly balanced system will have the flap angle curves flat and close to the neutral flap position (0˚) as in Figure 6.

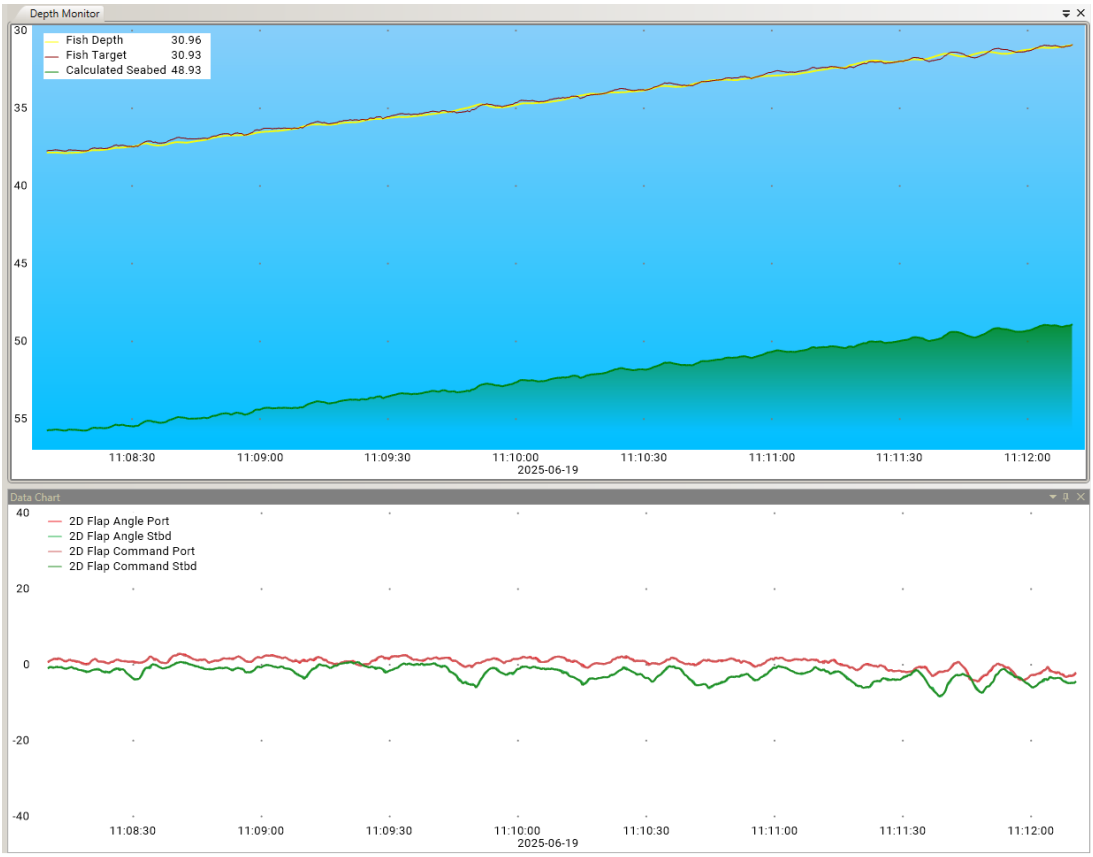

Figure 6 A well balanced system with flat flap angle curves around neutral position, comparing the Depth Monitor and Data Chart

In this scenario, the ViperFish needs to use minimal flap angle to stay at the desired depth, and only small flap adjustments for seabed topography changes. To have a perfectly balanced system the operator needs to find the right balance between the three factors that controls this, which are:

- Speed through water (STW)
- Cable length
- Survey depth

There are general guidelines in the ViperFish cable configuration document on EIVA's knowledge base, but it is up to the the operator to be able to analyse the behaviour and adjust accordingly.

In many cases there will be a given survey depth for the area, therefore it is only the two remaining factors the operator can change for system optimisation. Some of the sensors (Solstice/R2Sonic) may also be limited to a certain speed over ground to get the data densities required. Consequently, the operator is limited to adjusting the cable length to optimise the system performance – which also makes this a lot easier.

If the ViperFish is instructed to descend by the operator, or if seabed is changing to deeper depth, the motor flaps will respond by moving toward more negative angles than the starting point. If the ViperFish is instructed to ascend in the water column (either by the operator or shallower seabed), the flap motor angles will move toward more positive angles than the starting point.

Note: The ViperFish flaps angles are the opposite of the ScanFish

- Positive flap angle values cause the ViperFish to ascend
- Negative flap angle values cause the ViperFish to descend

3.1 Too much cable out, low STW and/or shallow depth

If you have a ViperFish system that is tracking a relatively flat seabed, and the flaps have balanced themselves on flat curves in the positive section of the **Data Chart** (the yellow zone in Figure 7), it should indicate to the operator that the ViperFish is constantly compensating for too much cable out, too low STW or shallow depth; it is actively try to ascend with the flaps.

In that case, the operator should consider recover some of the winch cable (at least if STW and survey depth are not changed) to move the curves from the yellow zone down to the green zone. In the green zone, the flaps are more efficient at adopting to seabed changes (both up and down) and give the system more manoeuvrability in general.

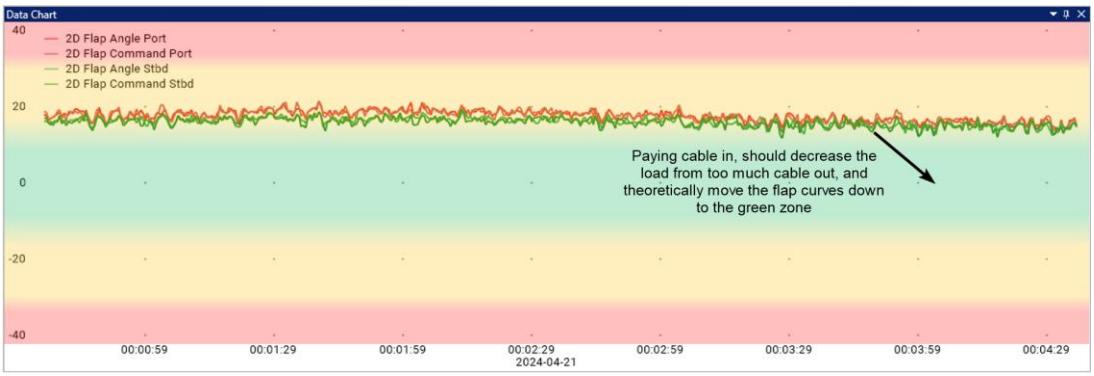

Figure 7 Data Chart with coloured zones for indicating good (green), medium (yellow) and bad (red) angle zones for the 2D flaps to be in

An alternative to paying cable in would be to increase the STW or go deeper with the ViperFish. That should both affect the system the same way, by lowering the flap angles of the 2D motors.

3.2 Too little cable out, high STW and/or deep depth

If you have the opposite situation where the flap curves have balanced themselves in the negative section as in Figure 8, the operator should consider either paying out additional winch cable, increase the STW or dive deeper to bring the flap angle curves from the yellow zone back to the green.

In the particular situation shown below, there is also a difference between starboard and port flap curves, which could either be a mechanical issue (asymmetry in the system or somehow damaged/affected flap motor) or environmental issue (cross current).

If it is mechanical, then the system should of course be recovered and fixed accordingly.

However, it is difficult to do anything from the operator's chair if it is due to cross currents.

Cross currents introduce some roll to the system that the flaps compensate for by being separated. In this case, the flap motor with highest angle (positive or negative) – in this case the port side motor – should be kept to a minimum.

A recommended approach to the situation in Figure 8 would be to pay additional cable out, that should give more positive flap angle to both motors, and bring both angles inside the green zone.

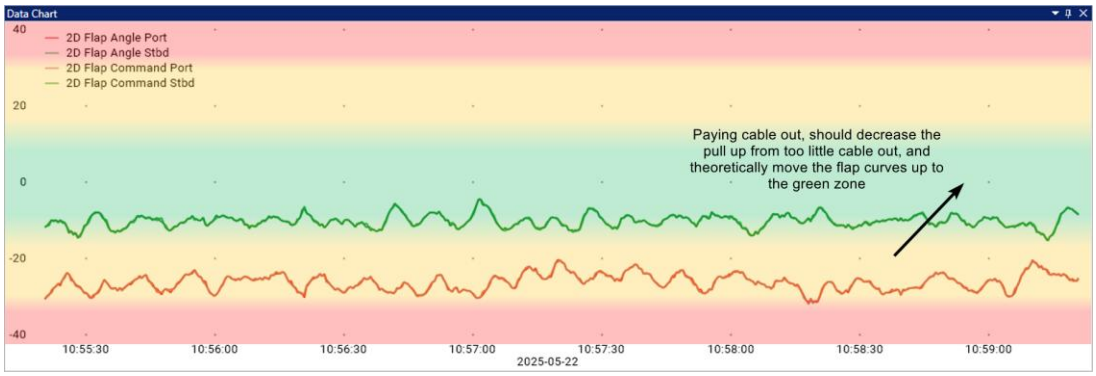

Figure 8 Data Chart with coloured zones for indicating good (green), medium (yellow) and bad (red) angle zones for the 2D flaps to be in

If working in a survey area where depths are constantly changing, the operator would need to make regular changes to the amount of winch cable out to give the ViperFish the best possible conditions for optimal performance.

4 Examples and actions

4.1 Balanced system

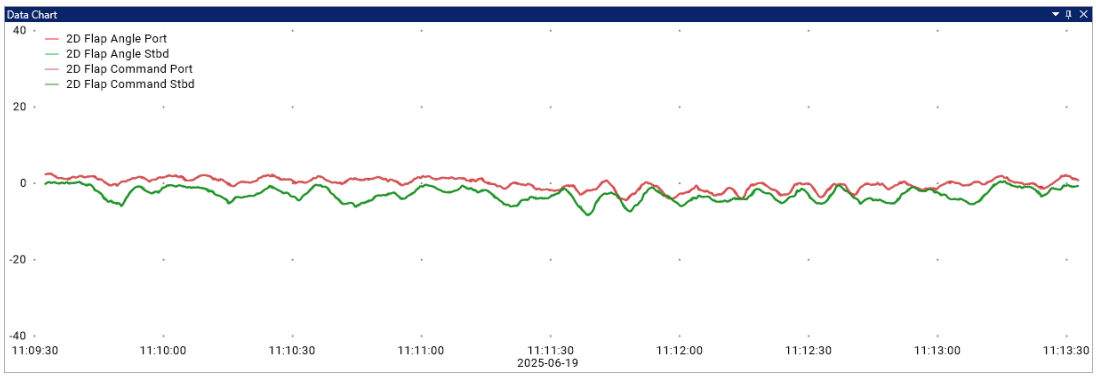

Figure 9 A well balanced system

An example of a well-balanced system. The curves are releatively flat, only adjusting to seabed changes and both are close to 0˚ of flap angle.

4.2 Flap locked out

Figure 10 A system that is suddenly not balanced – a turn that is too narrow

This incident is from a line turn, where the vessel makes a sharp turn, and suddenly the ViperFish doesn't have enough flow around the flaps to keep itself at the intended depth, so it starts to sink. Notice the **Fish depth** yellow line in the **Depth Monitor** panel, and as a consequence of the port flap being maxed out, the ViperFish picks up roll.

The flaps are going to positive degrees, meaning the ViperFish tries to ascend, which makes sense as there is a lack of STW to maintain the intended depth.

In this case the operator should get the vessel to straighten up to increase flow around the ViperFish flaps. This could be combined with paying winch cable in.

4.3 Port and starboard flap separation

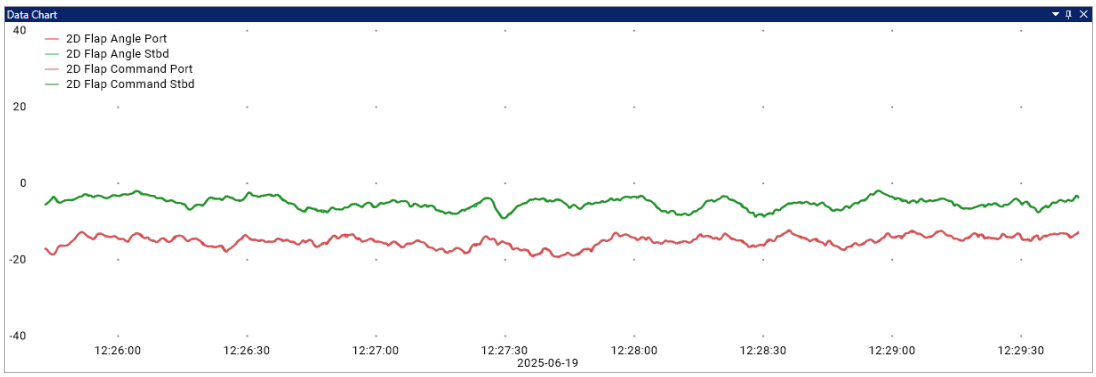

Figure 11 A reasonably balanced system with flap separation, but with consistent negative flap angles

An example of a reasonably balanced system. The angles indicate the ViperFish is constantly trying to descend to maintain the intended depth for both flaps. This will be due to one or more of the three things that affects the performance (cable length, STW and/or depth).

The curves are also slightly separated, which could be a result of side current that introduces some roll to the system – the flaps are compensating for it by being separated.

In this case, the ViperFish needs to be at the depth that it is in to perform the survey, so that cannot be changed. Therefore, either the cable length should be slightly longer or the STW should be slowed down – either could theoretically bring the flaps to more neutral angles.